REMARK R24: THE ALTERNATING-PRODUCT ARGUMENT OF THE CIRCULANT HADAMARD PROOF FOR NORM m, AND THREE PARALLEL APPLICATIONS (QUATERNARY, THREE-PHASE, KLEIN FOUR-GROUP)

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Abstract. OpenAI's preprint *The circulant Hadamard conjecture* (family 179 of `github.com/openai/math`) proves that an element $x \in \mathbb{Z}[i][C_{u^2}]$ with $xx^* = u^2$ has an alternating product of character values that is a root of unity of odd order, and combines this with a first-order 2-adic comparison. The proposition holds for every x with $xx^* = u^2$, and squareness is not used in it. We record that the same argument, with u^2 replaced by any m and the comparison done once, gives: no perfect quaternary sequence of length $2m$ ($m > 1$ odd); no perfect three-phase sequence of length $3m$ ($m > 1$, $3 \nmid m$); no perfect binary array on $C_2 \times C_2 \times C_{u^2}$ ($u > 1$ odd); and a rank-one identity for $x \in \mathbb{Z}[i][C_{pq}]$ with $xx^* = pq$. All four are checked in Lean 4 on top of the upstream formalization, using only the three standard axioms. **These are parallel generalizations**: the method and almost all of the proof are the preprint's. We claim no new idea.

1. Statements

Throughout, a function $a : G \to \mathbb{C}$ on a finite abelian group is *perfect* if $\sum_g a(g)\overline{a(g+s)} = 0$ for all $s \neq 0$. For $R \in \{\mathbb{Z}, \mathbb{Z}[i], \mathbb{Z}[\omega]\}$ and $N \geq 1$ write $R[C_N]$ for the group ring, x^* for the involution $g \mapsto g^{-1}$ combined with complex conjugation on R, and $J = \sum_g g$.

Proposition 1 (Prop′, Prop″)**.** *Let $N > 1$. Let $x \in \mathbb{Z}[i][C_N]$ with $xx^* = N$ and N odd, or $x \in \mathbb{Z}[\omega][C_N]$ with $xx^* = N$ and $3 \nmid N$. Then the alternating product $\Delta(x) = \prod_S x_S^{(-1)^{|S|}}$ of* [1, §3] *is a root of unity. For each prime $p \mid N$ its order is a power of p, so the order is odd, resp. prime to 3. If N has at least two distinct prime divisors, then $\Delta(x) = 1$.*

For $\mathbb{Z}[i]$ and $N = u^2$ this is [1, Prop. 3.4, `prop:alternating`]. The preprint itself says that the proposition "applies to every $x \in \mathbb{Z}[i][P]$ satisfying $xx^* = u^2$" (`introduction.tex` l. 191, `contradiction.tex` l. 3). Its proof uses the norm only through the facts that every character value has absolute value $\sqrt{N}$ and that N is a unit away from the primes dividing N. Both facts hold for any N. Over $\mathbb{Z}[\omega]$, the localization layer has to be redone at a prime above 3 with $\lambda = 1-\omega$, but the argument does not change.

Theorem 2 (Master Lemma, sum form)**.** *Let $(\mathcal{O}, \lambda, \ell) = (\mathbb{Z}[i], 1+i, 2)$ with $N > 1$ odd, or $(\mathbb{Z}[\omega], 1-\omega, 3)$ with $N > 1$, $3 \nmid N$. Let x_0 and x_j ($j \in s$) lie in $\mathcal{O}[C_N]$ with $xx^* = N$. Assume $x_j = x_0 + t_j y_j$ with $0 \neq t_j \in \lambda\mathcal{O}$, and $\sum_j y_j = eJ + \lambda W$ with $e \equiv c \pmod{\lambda}$, $c \in \mathbb{Z}$, $\ell \nmid c$. Then this is impossible.*

For $\mathbb{Z}[i]$, $N = u^2$, $|s| = 2$ this is the argument of [1, §4].

Theorem 3. (Q) *For odd $m > 1$ there is no perfect $a : \mathbb{Z}/2m \to \{\pm 1, \pm i\}$.*

(T3) *For $m > 1$, $3 \nmid m$, there is no perfect $a : \mathbb{Z}/3m \to \{1, \omega, \omega^2\}$.*

(T1) *For odd $N > 1$ there is no perfect $f : C_2 \times C_2 \times C_N \to \{\pm 1\}$. Equivalently, there is no Menon–Hadamard difference set in $C_2 \times C_2 \times C_{u^2}$, $u > 1$ odd. For non-square N this is trivial, since $(\sum f)^2 = 4N$.*

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(F) *Let $p \neq q$ be odd primes and $x \in \mathbb{Z}[i][C_{pq}]$ with $xx^* = pq$. Then $\varepsilon(x)\, x(g) = \pi_p(x)(g \bmod p) \cdot \pi_q(x)(g \bmod q)$ for all g. Here ε is the augmentation and π_p the push-forward to C_p. The statement is vacuous unless $p \equiv q \equiv 1 \pmod 4$, since $\varepsilon(x)\overline{\varepsilon(x)} = pq$.*

2. Why this is a parallel generalization

The new ingredient for each alphabet is a one-line normalization. The rest is the preprint's argument with u^2 replaced by m.

- (Q) Split $\mathbb{Z}/2m = C_2 \times C_m$ and write $h = H_0 + ZH_1$. Every element of $\{\pm 1, \pm i\}$ is $\equiv 1 \pmod{1+i}$, so $c = (H_0 + H_1)/(1+i)$ and $d = (H_0 - H_1)/(1+i)$ lie in $\mathbb{Z}[i][C_m]$ and have norm m. Moreover $d = c + (i-1)H_1$ with $H_1 \equiv J \pmod{1+i}$. Theorem 2 then applies with one index, $t = i - 1$ and $e = 1$. The preprint needs two comparisons; here one suffices.
- (T3) Write $h = H_0 + ZH_1 + Z^2H_2$ and $c_r = h(r)/(1-\omega)$ for $r \in \{1, \omega, \omega^2\}$. Then $c_\omega - c_1 = \lambda y_1$, $c_{\omega^2} - c_1 = \lambda y_2$ and $y_1 + y_2 \equiv 2\omega^2 J$. Apply Theorem 2 over $\mathbb{Z}[\omega]$ with $e = 2\omega^2$.
- (T1) Write $h = \sum X^a Y^b H_{ab}$ and $x_{\alpha\beta} = h(\alpha, \beta)/2 \in \mathbb{Z}[C_N]$, which have norm N. The three differences $x_{\alpha\beta} - x_{11} = -2b_j$ satisfy $\sum b_j \equiv J \pmod 2$. Apply Theorem 2 with three indices and $t = -2$.
- (F) By Prop$'$, $\Delta = 1$ for two primes. This reads $\hat{x}(1)\hat{x}(\chi\psi) = \hat{x}(\chi)\hat{x}(\psi)$ for primitive χ on C_p and ψ on C_q. Relabelling by units of $\mathbb{Z}/pq$ gives all character pairs, and Fourier inversion on $C_p \times C_q$ finishes the proof.

By the gate used in our project ("could an expert write this down in an afternoon from the original paper?"), the answer is yes. The 2-adic descent, the rigidity statements and the classification of admissible normalizations are not on the proof path of Q, T1, T3 or F, and they are not part of this remark.

3. Formal verification

The Lean files in `scripts/lean/OAI/LinearAlgebra/CirculantHadamard/` (`GeneralNorm/`: 7 files; `Eisenstein/`: 8 files; `RankOne/`: 2 files; 4085 lines) are added on top of `openai/math` commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. They do not modify any upstream file and import 15 unmodified upstream modules of the 179 formalization. The toolchain is Lean `v4.34.1` with the upstream `lake-manifest.json` (Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612`). `Checks.lean` prints the statements and axioms of Prop$'$, Prop$''$, both Master Lemmas, Q, T1 (both forms), T3 and F. Each depends only on `propext`, `Classical.choice` and `Quot.sound`. The equivalence of perfect binary arrays with Menon–Hadamard difference sets is standard and is not formalized.

4. Computational sanity checks

These are independent of the Lean development and share no code with the paper artifacts. They are unit tests only; the theorems rest on the proofs. Exhaustive search (normalized $a(0) = 1$) finds no perfect quaternary sequence of length $6, 10, 14$ and no perfect three-phase sequence of length $6, 12, 15$. As positive controls it finds $2, 8, 32$ quaternary sequences of length $2, 4, 8$ and $6, 54$ three-phase sequences of length $3, 9$. A meet-in-the-middle over the four normalized components finds no perfect binary array on $C_2 \times C_2 \times C_9$. On $C_2 \times C_2 \times C_3 \times C_3$ it finds 15552, which is consistent with the known $(36, 15, 6)$ Menon difference sets there [6]; one of them was checked directly.

5. What is not claimed

We claim no new method, and this remark makes no claim about individual lengths. Many lengths in these families are excluded by self-conjugacy [2, 4, 5], by field descent, or appear in the tables of [3]. For T1, Turyn's exponent bound already excludes $C_2 \times C_2 \times C_{u^2}$ whenever some prime

$p \mid u$, $p^a \parallel u$, is self-conjugate modulo $2(u/p^a)^2$. This covers all prime powers u and all odd $u < 39$. Theorem (F) is proved only for two distinct odd primes and $\mathbb{Z}[i]$. Correctness depends on the Lean kernel and on the faithfulness of the Lean statements; the reduction from difference sets to perfect arrays is not formalized. Related work by the same authors is in preparation.

Acknowledgement and AI disclosure

The method is OpenAI's [1]. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; independent adversarial reviews were run by separate AI sessions and are not human peer review.

References

[1] OpenAI, *The circulant Hadamard conjecture*, preprint, September 23, 2026, `https://github.com/openai/math`, commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, `preprints/The-circulant-Hadamard-conjecture-September-23-2026/`; Lean: `lean/OAI/LinearAlgebra/CirculantHadamard/`.

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[3] K. T. Arasu, Sequences and arrays with desirable correlation properties, in *Information Security, Coding Theory and Related Combinatorics*, IOS Press, 2011, 136–171.

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